

Final Report

Abstract

The project *Lego-ing Bacheng* is an interactive web-based narrative that explores the cultural erosion and memory in Bacheng's old town amidst rapid urban redevelopment. It merges immersive fieldwork findings – including photographs, audio, and oral histories – with creative storytelling and web technology to reanimate a disappearing local heritage. Through a multi-layered design, the project juxtaposes archival memories against present-day realities, inviting users to reflect on the tension between progress and preservation and to engage with the “spectral” traces of a vanishing community.

1. Introduction

This project—*Lego-ing Bacheng*—originated with a single photograph and a consequential misreading: a photograph of a tinplate house with 3 characters “方法科” (Fangfake, directly translated as Method Department) on it, with several men who seemed like construction workers sitting next to it. When we went to Bacheng, this place had already vanished in the trend of demolition. We spent several weeks attempting to locate the concrete office, site, or administrative unit of “方法科”. Finding no verifiable referent, we provisionally construed it as a government-instructed infrastructural and methodological department concerned with standardised urban construction—i.e., the coordinated demolition of older housing and the erection of new, uniform buildings under close supervision. That working definition accorded with conditions repeatedly observed in Bacheng: staged “old-street” façades, regimented rebuilds, inspection routines, and the steady north–south progression of clearance.

On revisiting the photograph, we noted that three additional characters follow “方法科”. We were unable to determine their content and, consequently, we treat *Fangfake* not as a fixed proper noun but as an analytic placeholder for a mobile apparatus of reform that is not site-specific yet leaves highly legible effects: demolition, displacement, and the retrofitting of vernacular environments to meet uniform standards. Our interview with a local restaurant

owner indicates that, within official discourse, such interventions are framed as the “correct” (正确的) form of cultural preservation: remove existing residents, regularise the built fabric, curate narratives for visitors, and certify the result. This position stands in tension with the lived practices and memories we documented on site.

Accordingly, the project does not attempt to resolve the literal identity of what follows “方法科”. Instead, it investigates the mechanisms and consequences associated with the phenomenon the term helped us to name. Also, our failure in searching for “方法科” conversely leads to our concept of cultural preservation in this specific area: you only see the value of preservation when you know there’s demolition happening everywhere simultaneously, what we are capturing is not snapshots that determines the historical value of objects through the very moment when the shutter blinks, but rather, durations of demolition that depicts the contrast before and after the changes, which, to us, is how something would be remembered, by revoking memories, not by fantacised legends. The research combines early-morning observations, photographic and audio documentation, and a semi-structured interview with a long-term resident to examine how redevelopment reshapes everyday life and memory in Bacheng. The final output—an interactive storytelling website—integrates these materials to juxtapose present conditions with recalled practices, and to analyse how “preservation” as implemented may undermine the very cultural vitality it claims to protect. The remainder of the paper presents background and conceptual grounding, details the creative and technical development of the site, and concludes with a discussion of contributions, limitations, and directions for future work.

2. Design and Creative Development

a. Narrative Design:

The narrative arc of our project mirrors our own fieldwork journey, transforming the user from a passive observer into an active participant in the search for meaning within Bacheng's transformation. It begins with an introduction to the spectral concept of “方法科”. The narrative tone is a blend of documentary observation and personal reflection. We present the facts and memories we collected, then layer them with our own critical commentary. This approach ensures the stories are rooted in authentic testimony while guiding the user toward the project's core philosophical questions. The user is left not with a resolved conclusion, but

with a felt understanding of the contrast between the government's "correct" method of preservation and the vibrant life that once defined the community

b. Visual and Spatial Design:

The photographic and audio materials collected during our fieldwork form the foundational layer of our visual narrative. We move beyond a simple gallery, treating each photograph as a portal to a deeper story. Our layout employs strategic juxtaposition and overlay to articulate this core theme. This creates a powerful, user-driven experience of erasure and change. The typography is clean and minimalist, ensuring the visual materials and personal narratives take center stage. To spatially ground the user in Bacheng's transformation, we have constructed a simplified 3D map of the town. This map is not photorealistic; instead, it is built from basic geometric 3D models, creating an abstract, almost architectural representation of the nine distinct zones we identified. This deliberate stylistic choice avoids a purely nostalgic recreation and instead presents Bacheng as a site of study - a diagram of change.

c. Interactive Component:

The website features two interactive modes to engage users actively with the themes of memory and reconstruction. Explore Mode centers on a navigable 3D map of Bacheng, where users discover floating models that unlock location-specific stories. Users are free to navigate this landscape, discovering floating 3D models that mark points of interest. Clicking on these models unlocks the deep storytelling for that specific location, presenting a curated fusion of photography, soundscapes, and narrative text. This free-form exploration mirrors the act of chasing the spectral “方法科”, transforming the user into an investigator of erasure and allowing them to piece together the non-linear narrative of cultural loss through their own curiosity. The narrative they piece together is not primarily one of architectural loss, but of vanishing human interactions. For instance, unlocking a point on the riverbank might reveal an AI-animated scene of children swimming and vendors lowering baskets from bridges. This design argues that the true culture of Bacheng did not reside in the static buildings, but in the lively, everyday rituals that once animated those spaces, and that these are what have been ultimately erased by progress.

In contrast, Tour Mode structures the user's journey through a conceptual wording game. This mode begins with the ambiguous photograph of the "方法科" (Fangfake) tin house. Users are tasked with deducing the three hidden characters that follow "方法科" by selecting words floating around. The system is designed to accept multiple valid combinations, with subtle hints available in the corner to guide the interaction without giving away the answers. Each correct word combination unlocks a specific thematic narrative page, such as the story of the Restaurant or the Bridge. This design creates a structured, episodic journey. Upon completing a story, the user is returned to the wording game to guess the next theme, progressively building a multifaceted understanding of Bacheng's transformation. This cyclical process of guessing, unlocking, and reflecting mirrors the investigative nature of our fieldwork. Finally, upon unlocking the last narrative, the game is complete, and the user graduates to the open-ended Explore Mode.

d. Sound or Multimedia Layer:

The soundscape of this project is exclusively sourced from our fieldwork, forming an authentic audio archive. We integrate two types of recordings: environmental sounds from key locations and direct audio excerpts from our interviewees. These sounds are not used as background filler but are paired with specific visual elements and narrative moments. This selective design serves a crucial conceptual purpose. By triggering the authentic sounds of a place while the user views its current state, we create a powerful, immersive dissonance.

3. Technical Implementation

a. Platform & Tools

Landing Page:

The website is developed with HTML/CSS/JavaScript to build its structure and basic UI elements, as well as [Three.js](#) to render the 3D scene in the map. The 3D models are built in Blender. The landing page consists of two layers of content: a 3D scene in the back, and textual introductions in the front. We control the layers by setting different “z-index” for each of them in CSS. When users scroll, the text scrolls down while the 3D scene zooms in at the 3D scene, which is achieved by setting the camera’s position on the Z axis according to scroll distance.

Explore Mode 3D Map Page:

The 3D map page consists of a main 3D map and 9 floating 3D icons, which can lead users to narrative pages. The 3D objects are made manually in Blender by using simple geometry shapes and minimal modifiers, keeping mesh numbers low, both in order to create a nostalgic game aesthetic and to reduce the renderer's workload. We exported .glb files of the models for the website to load in. Apart from the objects, we add ambient light and point lights to this scene. The animation of icons floating is achieved by changing their locations on the Y axis according to a sin function with respect to time (`const t = clock.getElapsedTime(); bricks.position.y = Math.sin(t)*0.6 + bricksOriginalY`). The interaction on this 3D map is indicated on the top left corner of this page: drag to rotate, scroll to zoom, Shift+drag to move, and click on the objects to redirect to other webpages for detailed narratives for this location. The navigation is achieved by the OrbitControls addon from [Three.js](#). Clicking for redirecting is achieved with Raycaster from [Three.js](#), which calculates if the mouse intersects with objects on a 2D screen.

Tour Mode with Character Game Page:

The character game is also made with HTML/CSS/JavaScript. We use a `Math.random()` function to randomly place Chinese characters across the screen. The `addEventListener()` method is used for most of the interactions, including dragging and dropping. The three characters dragged and dropped in the designated areas are stored in an array. When these characters equal a certain correct answer in the game, the website redirects them to a certain webpage (`window.location.href = "restaurant0.html"`) according to the narrative arc. When the user finishes viewing this certain webpage, they will again land in the character game and drag and drop a new combination of characters. Meanwhile, the background image will change according to the previously viewed page by keeping this information in `localStorage` (`const lastPage = localStorage.getItem("lastPage");`) When the user has seen every narrative page, they are redirected to the 3D map scene to start explore mode.

Narratives Page:

The interactive narrative website was built using a standard web development stack of HTML, CSS, and JavaScript, ensuring broad accessibility and compatibility across modern browsers. The initial development process was accelerated by using HaiSnap AI to generate a foundational website template that aligned with our desired aesthetic and layout. This

template was then extensively customized and refactored to integrate our project-specific assets, including photographs, audio recordings, and written narratives, and to implement the unique interactive features required for our storytelling. To create a dynamic and engaging user experience, we implemented a variety of interactive effects using custom JavaScript and libraries such as scroll-triggered animations, timed fade-ins and fade-outs, drag-and-drop functionality, image zoom, and infinite loops. Throughout the development process, AI-assisted programming tools like GitHub Copilot and ChatGPT were instrumental in polishing the code and ensuring the final website ran smoothly and efficiently across different devices.

Memory Layer Animation:

To visualize the past from interview transcripts while minimizing our own biased interpretation, we employed a mediated AI pipeline. We used DeepSeek to analyze the interviewee's own descriptions, focusing on generating prompts that were strictly descriptive and based solely on the provided testimony, for example, "children swimming in a canal alongside wooden boats" or "a vendor lowering a wicker basket from a stone bridge." This prompt engineering strategy was a deliberate ethical choice to minimize our own artistic bias and interpretive interference. The resulting prompts were executed by Hailuo AI to create the animations, which were then composited with contemporary photographs using TouchDesigner.

b. Web Integration & Accessibility

We obtained informed consent from the restaurateur for both participation and audio recording through a signed consent form, which specifies that the recording will be used solely for academic purposes within the context of this course. As our narratives incorporate excerpts from the interview audio, the website is not publicly accessible and is restricted to classroom use. Given that the interview may contain sensitive information related to the participant's personal opinions or demographic details, we have taken care to exclude such content from the published narratives. The website focuses exclusively on general stories about Bacheng that do not involve any identifiable personal information.

4. Discussion and Reflection

a. Evaluation of Outcomes

The project interprets the complex, spectral concept of“方法科” into a tangible and engaging user experience. The narrative clarity was a key strength, achieved by structuring the website around our actual fieldwork journey. This allowed users to follow a logical arc from initial curiosity about Fangfake to a deeper understanding of its impact on Bacheng's community and culture. User engagement was effectively fostered through the dual-mode interactivity: the freedom of Explore Mode catered to self-directed discovery, while the conceptual wording game in Tour Mode prompted personal reflection on the nature of cultural narratives. The atmosphere, built directly from our field-recorded sounds and photographs, proved highly effective in creating an immersive and emotionally resonant sense of place.

b. Limitations and Challenges

While the core interactive structure was realized, certain envisioned elements could not be fully implemented due to technical and time constraints. Most notably, the 3D map's visual style had to be simplified. We originally intended to apply a toon shader to create a pseudo-2D, illustrative effect that would better align with the project's conceptual and poetic tone, but this was deprioritized to ensure the map's core navigational functionality was robust. Furthermore, the creation of the generative AI vignettes for the memories remained a conceptual prototype rather than a fully integrated feature. This limitation was shaped by both the technical complexity of producing and compositing high-quality animations and ongoing ethical considerations regarding the accurate and respectful representation of oral testimony. Also, the intentional diversity in page design and interactive methods, while aimed at making exploration more dynamic and engaging, risked creating a feeling of incoherence across the narrative. This was a conscious trade-off where we prioritized unique, place-specific storytelling over a perfectly uniform user interface. Second, due to time limits, some interactive pages lacked detailed user guidance. This might leave users uncertain about how to engage with certain features, potentially hindering the full immersive experience we intended. Finally, the project's narrative, while deep, is primarily built around a single, albeit profoundly insightful, interview. A broader range of resident voices would have provided even more nuanced perspectives on the changes in Bacheng.

5. Future Directions

This website could potentially integrate AI into the character game, with AI interpreting different permutations of the three characters and generating an image or animation based on Bacheng's cultural elements that we collected for this project.

6. Conclusion:

Lego-ing Bacheng transforms a failed search for the vanished “方法科” into an inquiry into how cultural preservation and erasure intertwine within urban redevelopment. What began as documentation evolved into an interactive narrative that reanimates the fading atmospheres of Bacheng through sound, image, and user participation. By merging ethnographic observation with web-based storytelling, the project invites users to experience disappearance not as distant history but as a lived, affective, and progressive process. Its key contribution lies in demonstrating how digital interactivity can serve as both research and remembrance, where the user’s actions mirror the work of reconstruction itself. Each gesture of navigation or annotation becomes a form of witnessing, linking personal reflection to collective memory. The project’s fragmentary design intentionally echoes Bacheng’s own incompleteness, suggesting that preservation today must move beyond static restoration toward active engagement with loss. In allowing users to build and unbuild memory piece by piece, *Lego-ing Bacheng* proposes that remembering is itself a creative, unfinished act.

Appendix

Box folder to download our project file:

<https://duke.box.com/s/rbt1idxl7ec872u9mefw0kw3rowfw8i>

For privacy and financial reasons, we have not published the website now, so please download our project folder from Duke Box for preview. The project folder by itself cannot render 3D scenes locally, to run it locally please reference the [three.js](#) documentation to install [Node.js](#) and Vite on the computer first.

<https://threejs.org/manual/#en/installation>

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